

Project Report

Project: Voltage-Controlled Phase Shifter

Analog Integrated Electronics Laboratory

Item	Details
Project	Voltage-Controlled Positive Phase Shifter
Main circuit type	First-order active all-pass phase shifter
Implementation	VCVS-based active stage + NMOS voltage-controlled resistor
Technology device used	gpd90 nmos1v
Final design frequency used for phase table	10 kHz
Final note	Supply-powered op-amp was attempted but could not be simulated because of Cadence AHDL compile error

1. Abstract

This project designs and simulates a voltage-controlled phase shifter. The target was to obtain a phase shift from 0 degree to 90 degree using a control voltage from 0 V to 1 V over the given frequency range of 1 kHz to 100 kHz. The main design was made using a first-order active all-pass filter. A VCVS was used as the active amplifier and an NMOS transistor from gpd90 was used as a voltage-controlled resistor.

The project was developed in stages. First, a fixed-resistor all-pass phase shifter was built to understand the basic phase-shifting behavior. Then the fixed resistor was replaced by an NMOS so that the phase could be controlled by a voltage. At first the circuit produced negative phase shift, meaning the output lagged the input. After that, the RC branch was changed to obtain a positive phase shift, so the output leads the input. The final positive phase-shifter achieved approximately 1.18 degree at $V_{ctrl} = 0$ V and approximately 90.66 degree at $V_{ctrl} = 1$ V at 10 kHz.

A supply-powered op-amp was also attempted using the ahdlLib opamp model. However, Cadence/Spectre could not compile the op-amp Verilog-A/AHDL model because of a tool environment error. Therefore, the final working simulations were completed using the VCVS model. For this reason, the phase-control behavior is verified, but the 2 V to 5 V supply voltage requirement was not fully simulated in the final VCVS version.

2. Project Specifications

Parameter	Required value	Unit	Status in this work
Frequency range	1 - 100	kHz	AC sweep performed from 1 kHz to 100 kHz
Phase shift range	0 - 90	degree	Achieved as positive phase shift at 10 kHz
Control voltage	0 - 1	V	Swept using vctrl from 0 V to 1 V
Supply voltage	2 - 5	V	Not fully verified with VCVS; supply-op-amp attempt failed due Cadence AHDL compile issue

3. Basic Idea of a Phase Shifter

A phase shifter changes the timing position of a signal without changing its frequency. If the input is a sine wave, the output is another sine wave with the same frequency but shifted in phase.

$$V_{in} = A \sin(\omega t)$$

$$V_{out} = A \sin(\omega t + \phi)$$

Here, ϕ is the phase shift. If $\phi = 0$ degree, input and output are in phase. If $\phi = +90$ degree, the output leads the input by one quarter of a cycle. At 10 kHz, one full period is 100 microseconds. Therefore, a 90 degree shift is 25 microseconds.

$$T = 1/f = 1/10 \text{ kHz} = 100 \text{ us}$$

$$90 \text{ degree time shift} = T/4 = 25 \text{ us}$$

4. Why an All-Pass Filter Was Used

A simple RC circuit can shift phase, but it also changes the signal amplitude strongly. For this project, we wanted the gain to stay close to 0 dB while the phase changes. That is why an all-pass filter was selected.

An all-pass filter ideally has constant magnitude:

$$|H(j \omega)| = 1$$

In dB, this means:

$$20 \log_{10} |H(j \omega)| = 0 \text{ dB}$$

The gain plot later shows that the gain variation is only in milli-dB, so the amplitude is almost constant. This confirms the all-pass behavior.

5. Theory and Equations Used

5.1 Negative-phase all-pass version

The first circuit we built was the common first-order active all-pass phase shifter. Its RC branch had the resistor first and the capacitor to ground. This produced a negative phase shift, meaning the output lagged the input.

$$H(s) = (1 - sRC) / (1 + sRC)$$

$$\phi = -2 \tan^{-1}(\omega RC)$$

This circuit was useful for learning and debugging, but the project requirement was interpreted as needing positive phase from 0 degree to +90 degree. Therefore, we later changed the circuit topology.

5.2 Positive-phase all-pass version

To get actual positive phase shift, the capacitor was placed in series between V_{in} and V_p , and the resistance path was connected from V_p to ground. This gives a leading phase response.

$$H(s) = (sRC - 1) / (sRC + 1)$$

$$\phi = 180 \text{ degree} - 2 \tan^{-1}(\omega RC)$$

For this equation, if R is very large, ωRC is large and the phase approaches 0 degree. If R is chosen so that $\omega RC = 1$, the phase becomes approximately 90 degree.

$$\text{For } \phi = 90 \text{ degree: } \omega RC = 1$$

$$R = 1 / (2 \pi f C)$$

5.3 Design calculation at 10 kHz

The control table and transient verification were taken at 10 kHz because it is inside the specified range of 1 kHz to 100 kHz and is convenient for seeing a 100 microsecond period in transient simulation.

$$C = 100 \text{ pF}$$

$$f = 10 \text{ kHz}$$

$$R_{\text{target}} = 1 / (2 \pi \times 10 \text{ kHz} \times 100 \text{ pF})$$

$$R_{\text{target}} = 159.15 \text{ k ohm}$$

This is why the resistance around the V_p node was designed to become approximately 159 k ohm when $V_{\text{ctrl}} = 1 \text{ V}$ in the positive-phase design. At this point, the phase becomes close to +90 degree at 10 kHz.

5.4 NMOS as a voltage-controlled resistor

The NMOS transistor was used as a voltage-controlled resistor. In the triode region, the approximate on-resistance of an NMOS is:

$$R_{on} \approx 1 / [\mu_n C_{ox} (W/L) (V_{GS} - V_{TH})]$$

This equation shows two important things:

- Increasing V_{ctrl} increases V_{GS} and reduces NMOS resistance.
- Increasing W reduces resistance.
- Increasing L increases resistance.
- The resistance is not perfectly linear with V_{ctrl} , so the phase-vs-control curve is also nonlinear.

Because a real NMOS is nonlinear and the model parameters are hidden inside the gpdk90 model, exact W and L cannot be found from a simple hand formula alone. We used theory to estimate values and then did simulation-based tuning to settle in the best working region.

6. Fixed-Resistor Verification

Before using the NMOS, a fixed resistor version was built. This was done to check whether the all-pass phase shifter itself was connected correctly. For the first fixed test, $R_{var} = 159 \text{ k ohm}$ and $C = 100 \text{ pF}$ were used.

Component	Value	Reason
R1	100 k ohm	Chosen equal to R2 for all-pass operation
R2	100 k ohm	Feedback resistor; equal to R1
Rvar	159 k ohm	Gives 90 degree phase shift at 10 kHz with $C = 100 \text{ pF}$
C0	100 pF	Gives practical resistor values for 10 kHz design point
VCVS gain	100 k	High enough to act like an ideal op-amp for this lab

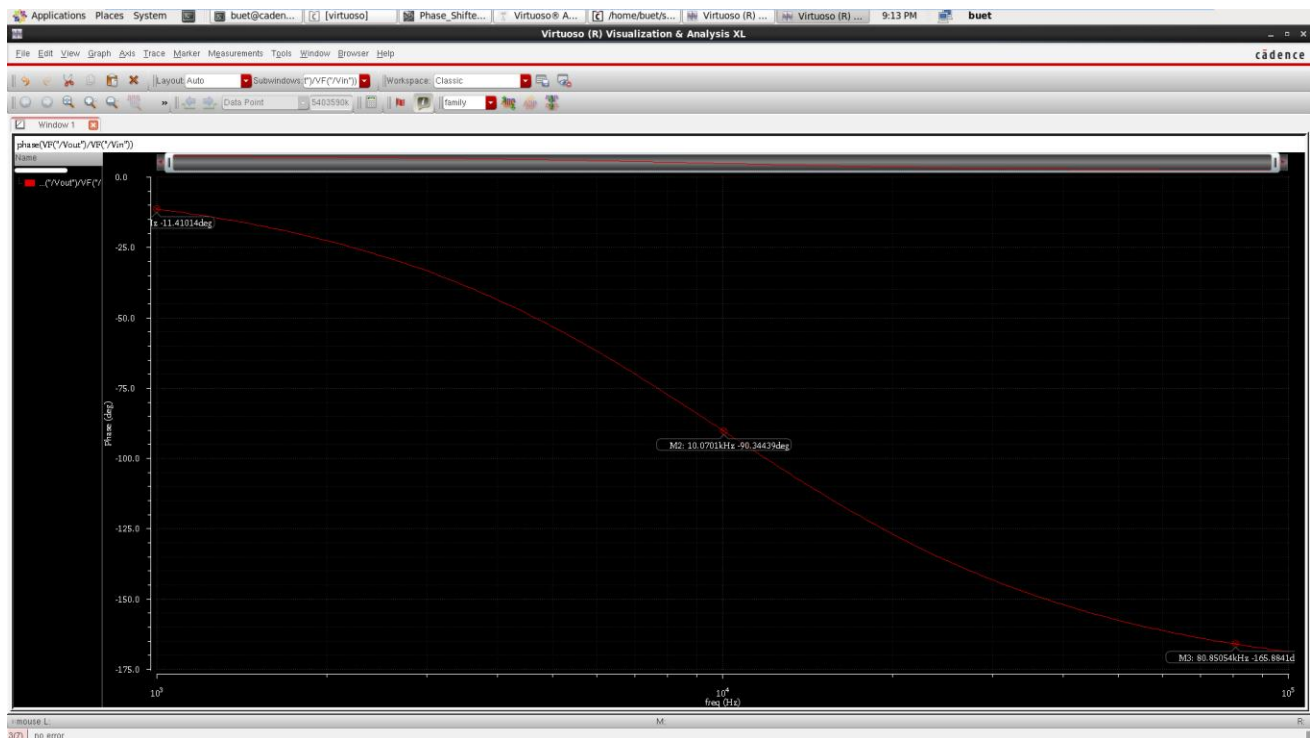


Figure 1: Fixed-resistor phase plot showing about 90 degree at 10 kHz

The fixed-resistor simulation gave about 90 degree phase shift near 10 kHz, which matched the theoretical calculation. This confirmed that the all-pass structure was correct.

7. Negative Phase Version with NMOS

After the fixed-resistor test, R_{var} was replaced by an NMOS. The first NMOS version used the lagging all-pass topology. In that circuit, the phase was negative because V_{out} lagged V_{in} . This was not wrong electrically, but it was not the final target because we wanted positive phase.

The NMOS-only version initially gave very small phase because the NMOS resistance was too low. The W/L values were changed to increase resistance. From the fixed-resistor result, we knew that about 159 k ohm was needed for 90 degree at 10 kHz. Simulated phase values were used to estimate the effective NMOS resistance and then scale L/W.

The negative phase circuit eventually achieved approximately -90 degree to 0 degree using a parallel limiting resistor. This helped us understand the control behavior before converting the circuit to positive phase.

Quantity	Value used in negative-phase stage	Purpose
NMOS	gpd90 nmos1v	Voltage-controlled resistor
W	3.5 um in the final negative-phase tuning	Made R_{on} small enough at $V_{ctrl} = 1$ V
L	1 um	Longer than minimum for more resistive behavior
R_{lim}	163 k ohm	Limited maximum resistance so phase did not go beyond about 90 degree
C0	100 pF	Same design capacitor

8. Final Positive Phase Circuit

The final circuit was changed so that the output leads the input. The key change was in the V_p branch. Instead of placing the capacitor from V_p to ground, the capacitor was placed between V_{in} and V_p . Then the resistance path from V_p to ground was controlled by R_{series} and the NMOS.

The final positive phase branch is:

$$V_{in} \rightarrow C0 \rightarrow V_p$$

$$V_p \rightarrow R_{series} \rightarrow NMOS \rightarrow ground$$

$$V_p \rightarrow R_{bias} \rightarrow ground$$

R_{bias} is very large, so it does not strongly affect the AC phase when the NMOS is on. Its main job is to prevent the V_p node from floating when the NMOS is off.

8.1 Final component values

Component / parameter	Final value	Why this value was used
R1	100 k ohm	Input resistor at inverting node. Equal to R2 for all-pass behavior.
R2	100 k ohm	Feedback resistor. Equal to R1 to keep near-unity all-pass gain.
C0	100 pF	Selected so that the 90 degree resistance at 10 kHz is about 159 k ohm.
R_{series}	153 kilo-ohm	Chosen so that $R_{series} + NMOS$ on-resistance becomes close to 159 kilo-ohm at $V_{ctrl} = 1$ V.
R_{bias}	100e6 ohm	Very large DC path from V_p to ground; prevents floating node but has little AC effect.
NMOS type	gpd90 nmos1v	Available transistor model used as voltage-controlled resistor.

Component / parameter	Final value	Why this value was used
NMOS W	1.2 um	Reduced from larger width to reduce off leakage and make Vctrl = 0 closer to 0 degree.
NMOS L	1 um	Longer channel for more resistor-like behavior and controlled Ron.
VCVS gain	100 k	Used as high-gain active stage because ahdLib opamp could not compile.
Vctrl	0 V to 1 V	Gate control voltage for NMOS.

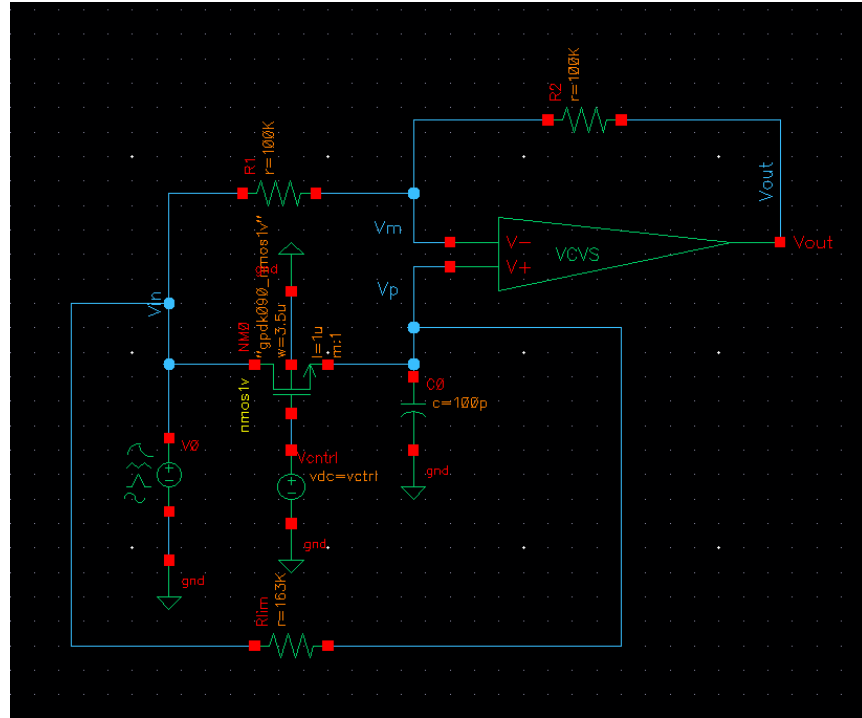


Figure 2: Final positive phase-shifter schematic

8.2 Why Rseries was around 153 k ohm

For 90 degree at 10 kHz, the total resistance seen in the Vp branch should be about 159.15 k ohm. When Vctrl = 1 V, the NMOS is on and it adds a small resistance in series with Rseries. Therefore:

$$R_{total} \approx R_{series} + R_{on,NMOS}$$

$$R_{total} \text{ should be close to } 159.15 \text{ k ohm}$$

The NMOS on-resistance depends on W, L, and Vctrl. After tuning, W = 1.2 um and L = 1 um gave a good balance. Rseries = 153 kilo-ohm then gave a final phase of about 90.66 degree at Vctrl = 1 V. This is very close to the required 90 degree.

8.3 Why Rbias was 100e6 ohm

When Vctrl = 0 V, the NMOS is off. Without Rbias, the Vp node can become floating in DC analysis because the capacitor blocks DC. A floating node can cause simulation errors or unstable results. Rbias provides a DC path to ground.

Rbias was chosen as 100e6 ohm because it is very large. A large value gives a DC path but does not significantly load the AC circuit. Therefore, it keeps the circuit stable without destroying the phase-shifting action.

9. Cadence Implementation Steps

1. Build the VCVS all-pass structure with $R1 = R2 = 100 \text{ k ohm}$.
2. Verify the fixed-resistor version first using $Rvar = 159 \text{ k ohm}$ and $C0 = 100 \text{ pF}$.
3. Replace the fixed resistor by an NMOS and add the Vctrl voltage source at the NMOS gate.
4. Observe that the first NMOS version gives negative phase shift. Use it to understand tuning and NMOS resistance behavior.
5. Change the Vp branch to the positive-phase topology: $Vin \rightarrow C0 \rightarrow Vp$ and $Vp \rightarrow Rseries \rightarrow NMOS \rightarrow$ ground.
6. Add $Rbias = 100e6 \text{ ohm}$ from Vp to ground to avoid floating node.
7. Run AC analysis from 1 kHz to 100 kHz.
8. Sweep Vctrl from 0 V to 1 V and observe the phase at 10 kHz.
9. Run gain plot to confirm that the circuit is almost all-pass.
10. Run transient simulation at $Vctrl = 0 \text{ V}$ and $Vctrl = 1 \text{ V}$ to verify time-domain behavior.

10. Simulation Settings

Simulation item	Setting used
AC analysis frequency range	1 kHz to 100 kHz
AC input source	V0 with AC magnitude = 1 and DC = 0
Control voltage source	Vctrl source with DC value = vctrl
Parametric sweep	vctrl = 0 V to 1 V in 0.1 V steps
Transient input signal	Sine wave, 10 kHz, 20 mV peak, DC level = 0
Transient stop time	500 us
Model library	gpd090.scs, NN section
Output phase expression	$\text{phase}(VF("/Vout") / VF("/Vin"))$
Gain expression	$\text{dB20}(VF("/Vout") / VF("/Vin"))$

11. Results and Discussion

11.1 Phase control result at 10 kHz

The following table shows the phase shift at 10 kHz for different control voltages. These values were taken from the final positive-phase circuit.

Vctrl (V)	Phase shift at 10 kHz (degree)
0.0	1.18047
0.1	19.13928
0.2	68.0335
0.3	83.528
0.4	87.347
0.5	88.825
0.6	89.42781
0.7	89.50859
0.8	90.2587
0.9	90.372
1.0	90.657

The phase starts close to 0 degree at $V_{ctrl} = 0$ V and reaches close to 90 degree at $V_{ctrl} = 1$ V. This meets the main phase-shift requirement of the project.

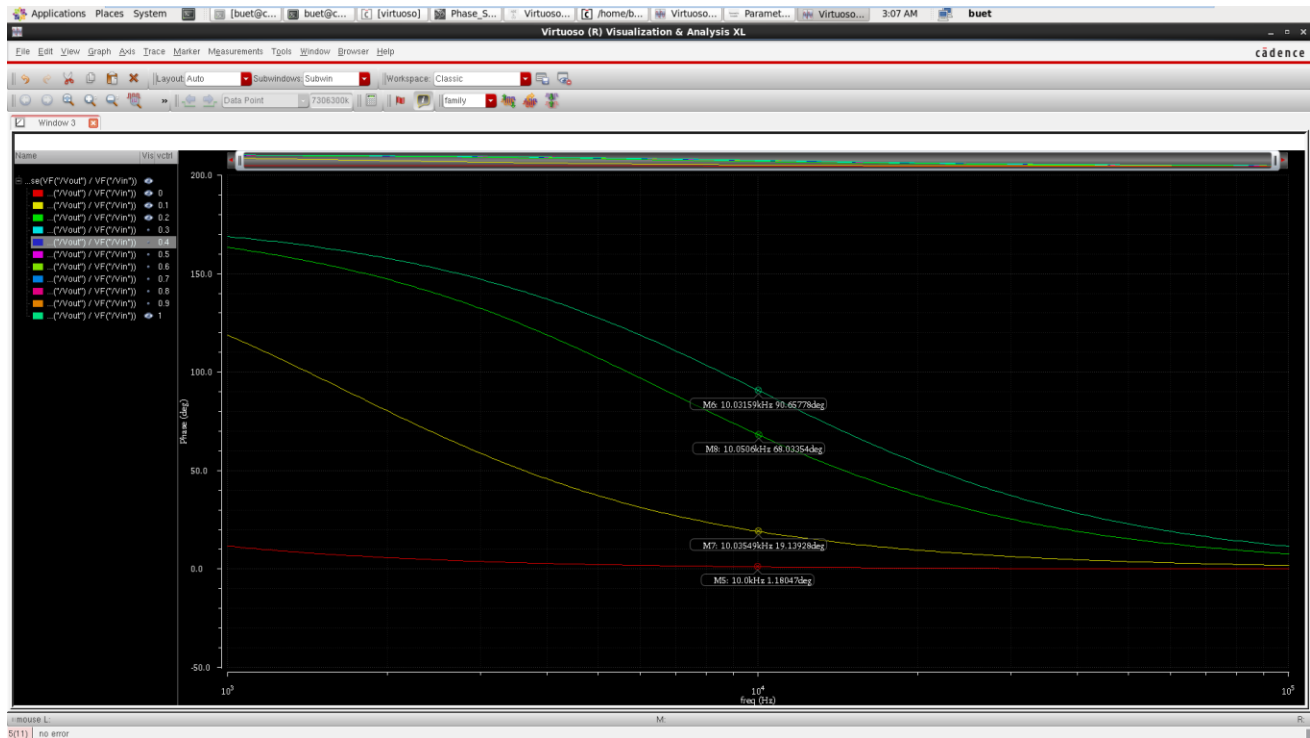


Figure 3: Final AC phase sweep for $V_{ctrl} = 0$ V to 1 V

11.2 Why the phase saturates after about 0.5 V

The phase does not increase linearly with V_{ctrl} . After about 0.5 V, the phase becomes almost saturated near 90 degree. This is expected because the NMOS enters a strong-on condition. At that point, the NMOS resistance becomes small compared to R_{series} .

$$R_{total} \approx R_{series} + R_{NMOS}$$

When R_{NMOS} becomes very small, changing V_{ctrl} further does not change R_{total} much. Therefore, the phase also changes very little. This is acceptable because the project requires the phase range, not a perfectly linear phase-vs-control response.

11.3 Gain result

The gain plot was taken using $\text{dB}20(VF("V_{out}") / VF("V_{in}"))$. The gain variation was very small and shown in milli-dB. This means the gain is practically close to 0 dB. Therefore, the circuit behaves like an all-pass filter: it mainly changes phase while keeping amplitude almost constant.

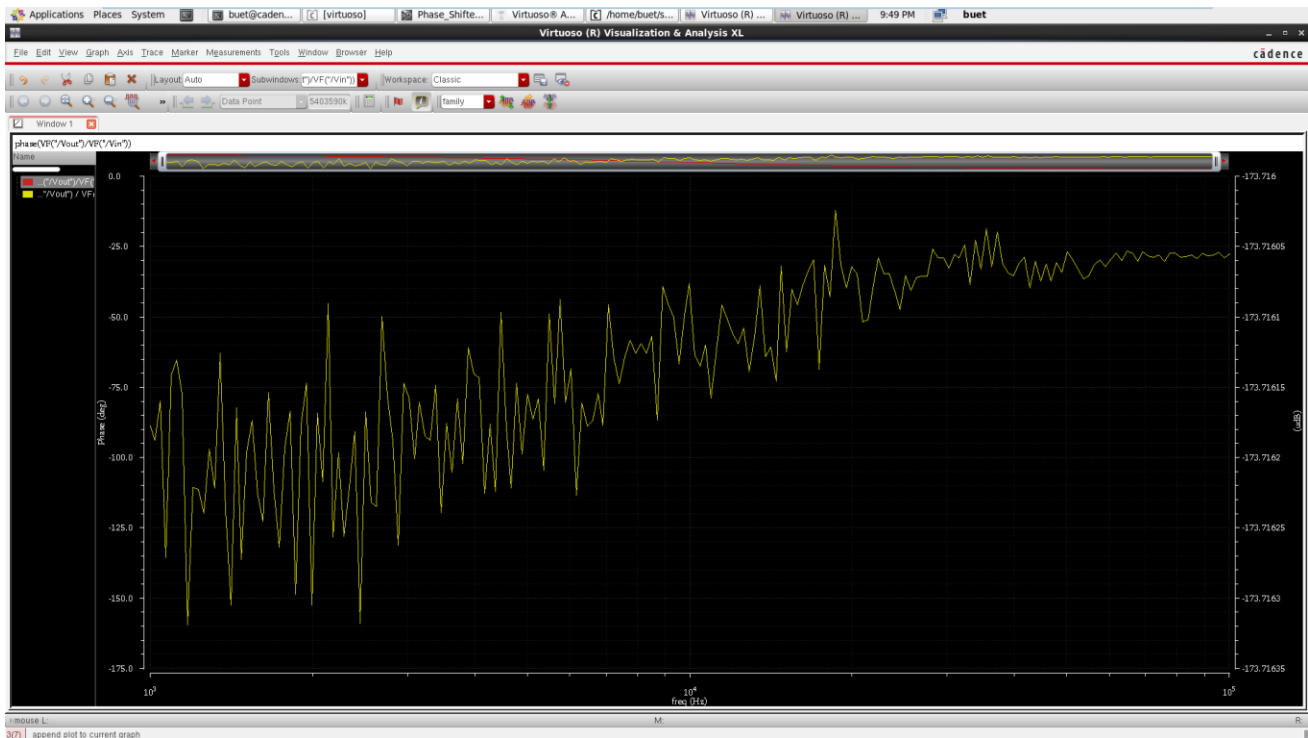


Figure 4: Gain plot showing nearly 0 dB response

11.4 Transient result for $V_{ctrl} = 0$ V

For $V_{ctrl} = 0$ V, the AC phase shift was about 1.18 degree. In time domain, this is a very small delay:

$$\Delta t = (1.18 / 360) \times 100 \text{ us} \approx 0.33 \text{ us}$$

Therefore, V_{in} and V_{out} almost overlap in the transient plot. This is what was observed.

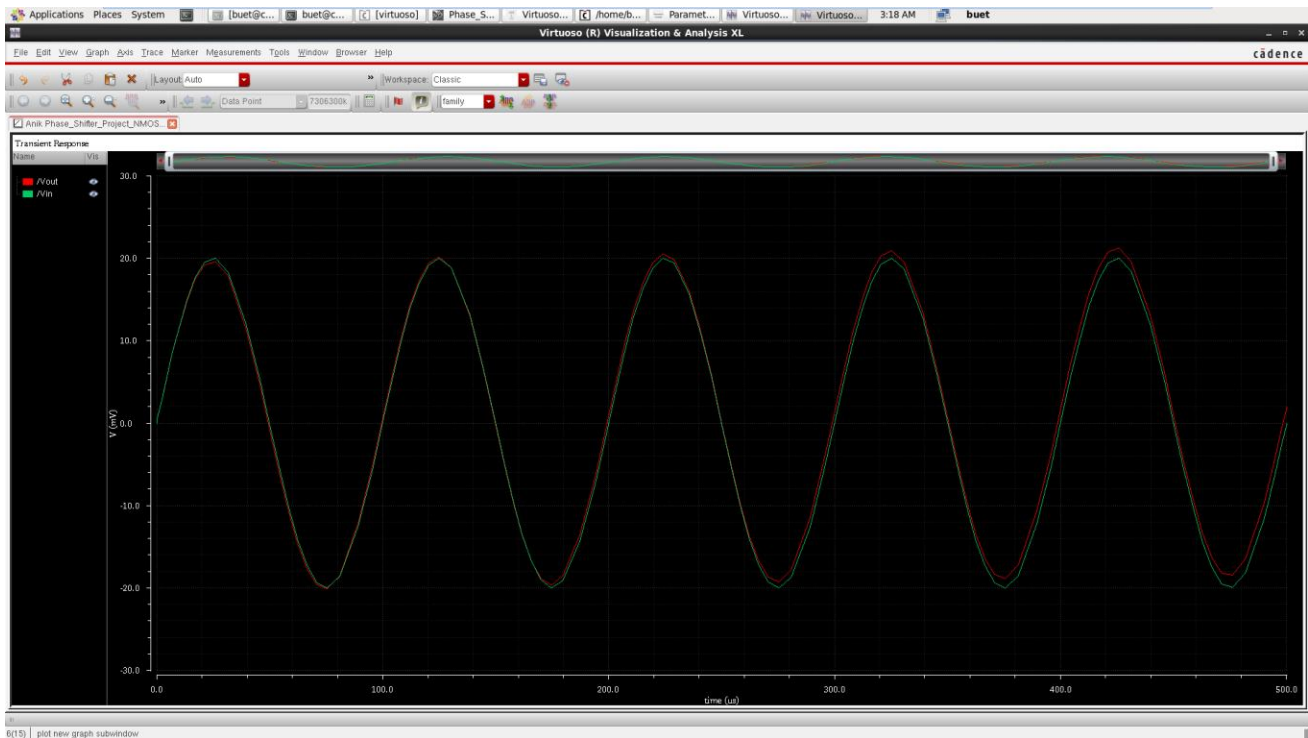


Figure 5: Transient plot at $V_{ctrl} = 0$ V showing V_{in} and V_{out} almost overlapping

11.5 Transient result for $V_{ctrl} = 1\text{ V}$

For $V_{ctrl} = 1\text{ V}$, the phase shift was close to 90 degree. At 10 kHz, the period is 100 μs , so 90 degree corresponds to 25 μs .

$$\Delta t = (90 / 360) \times 100\text{ }\mu\text{s} = 25\text{ }\mu\text{s}$$

In the transient plot, V_{out} leads V_{in} by about one quarter of a cycle. This confirms the positive 90 degree phase-shift behavior.

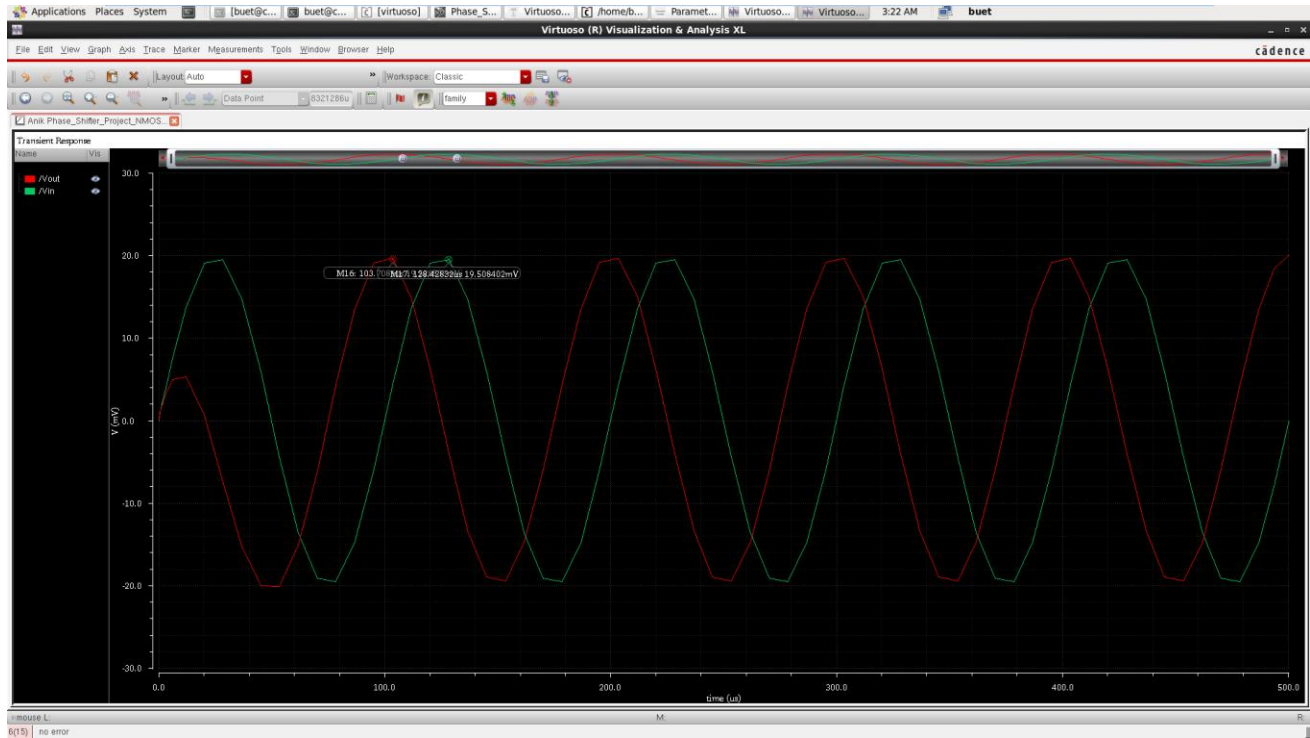


Figure 6: Transient plot at $V_{ctrl} = 1\text{ V}$ showing about 90 degree lead

11.6 Why the first red curve was smaller in the 90 degree transient plot

In the $V_{ctrl} = 1\text{ V}$ transient plot, the first red output waveform peak was smaller than the green input waveform. This happened because of start-up transient. At the start of transient simulation, the capacitor has an initial condition, usually zero. But for a true 90 degree lead, the output should already be near a peak when the input starts at zero. Since the simulator starts from initial conditions, the first cycle is not yet in steady state.

After the first cycle, the circuit settles and the output amplitude becomes close to the input amplitude. Therefore, phase and amplitude should be measured from later cycles, not from the first cycle.

12. Supply Voltage Requirement and Op-Amp Issue

The project specification included a supply voltage range of 2 V to 5 V. In a real design, this means the active amplifier should have supply pins such as VDD and VSS. However, the main working circuit used a VCVS. A VCVS is an ideal voltage-controlled voltage source and has no supply pins. Therefore, the VCVS version does not explicitly verify the 2 V to 5 V supply requirement.

To include the supply voltage requirement, an ahdLib opamp was added. The intended connections were: V_p to the positive input, V_m to the negative input, output to V_{out} , positive supply to VDD, negative supply to VSS, and v_{ref} to ground. Op-amp properties such as gain = 100k and unity gain frequency = 100e6 were used.

ERROR (VACOMP-1008): Cannot compile ahdlcmi module library.
ERROR (SFE-91): Error when elaborating the instance opamp.

```

input scs +=scsbars +log ../psf/spectre.out +intermagso \
+aggressivenesspectral 7550 1 -fscmat psfsl -rsw ../psf \
+lightcount 900 -narrow 5 -narrow 5

spectre pid = 8174

Loading /home/buut/cadence/MSMTI1/tools lr0b6/cmi/lib/S/0/libbifirnon_sh.so ...
Loading /home/buut/cadence/MSMTI1/tools lr0b6/cmi/lib/S/0/libbipilips_5_sh.so ...
Loading /home/buut/cadence/MSMTI1/tools lr0b6/cmi/lib/S/0/libbipilips_sh.so ...
Loading /home/buut/cadence/MSMTI1/tools lr0b6/cmi/lib/S/0/libbiprwn_2h.so ...
Loading /home/buut/cadence/MSMTI1/tools lr0b6/cmi/lib/S/0/libbmodels_sh.so ...
Reading file: /home/buut/cadence/Phase_Shifter_Project/UMOS_VCD/spectre/schematic/netlist/input.scs
Reading file: /home/buut/cadence/gp-d090 v4 6/models/spectre/gp-d090.scs
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Reading link: /home/buut/cadence/MSMTI1/tools lr0b6/spectre/etc/abdl/discriminate_h
Reading file: /home/buut/cadence/MSMTI1/tools lr0b6/spectre/etc/abdl/discriminate_vmas
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Reading file: /home/buut/cadence/MSMTI1/tools lr0b6/spectre/etc/abdl/constants_vmas
Reading file: /home/buut/cadence/gp-d090 v4 6/models/spectre/roaioe_va_va
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Reading file: /home/buut/cadence/gp-d090 v4 6/models/spectre/gp-d090_ipulac.scs
Reading link: /home/buut/cadence/15615/tools/dff1/samples/artist/shdlib/opamp/VerilogA/opamp.va
Reading link: /home/buut/cadence/15615/tools
Reading file: /home/buut/cadence/75510/tools lr0b6/dff1/samples/artist/spectreRML/Verilog-A/analog/opamp.va

Time for NBR Parsing: CPU = 305.963 ns, elapsed = 295.188 ns.
Time accumulated: CPU = 305.963 ns, elapsed = 295.188 ns.
Peak resident memory used = 12.4 Mbytes

Created directory input shdlibRML/ (775)
Created directory input shdlibRML/4330_artist_shdlib_opamp_verilogA_verilogA.va.opamp.shdlibcml/ (775)
Created directory input shdlibRML/4330_artist_shdlib_opamp_verilogA_verilogA.va.opamp.shdlibcml/limux/ (775)
Compiling shdlibcml module library

Error found by spectre during ANEL read-in:
ERROR (WSE-900-1000) Cannot compile shdlibcml module library. Check the log file input shdlibRML/4330_artist_shdlib_opamp_verilogA_verilogA.va.opamp.shdlibcml/limux/ for details. If the compiler ran out of memory, use 'setenv CDS_CHI_COMPLEXLEVEL 0', and try again.
ERROR (WSE-900-1000) Error when elaborating the instance opamp. Simulation should be terminated.

Time for Elaboration: CPU = 76.969 ns, elapsed = 190.648 ns.
Time accumulated: CPU = 304.941 ns, elapsed = 486.951 ns.
Peak resident memory used = 38.6 Mbytes

spectre terminated prematurely due to fatal error.

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13. Issues Faced During the Project

Issue	What happened	How it was handled
Understanding Vin, Vp, Vm	At first the node meanings were confusing.	Vin was defined as input source node, Vp as non-inverting phase network node, and Vm as inverting feedback node.
Wrong Vp/Vm short	In early schematic, Vp and Vm were accidentally connected.	Separated Vp and Vm. Vm kept only for R1, R2, and VCVS negative input.
Transient did not run	Cadence said cell was modified since last extraction.	Used Check and Save before running simulation.
Wrong ADE design state	ADE tried to simulate an old cell.	Closed ADE and launched ADE from the correct schematic.
Negative phase result	First all-pass circuit produced negative phase.	Changed the RC branch to positive-phase topology.
Vctrl response saturation	Phase became almost constant above about 0.5 V.	Explained as NMOS strong-on behavior where Rseries dominates total resistance.
Supply op-amp compile failure	ahdlLib opamp failed with AHDL compile error.	Returned to VCVS implementation and documented the issue honestly.

14. Final Verification Against Requirements

Requirement	Final observation	Result
Frequency range 1-100 kHz	AC sweep was performed over 1 kHz to 100 kHz.	Verified in simulation
Phase shift 0-90 degree	At 10 kHz, phase varied from about 1.18 degree to 90.66 degree.	Satisfied
Control voltage 0-1 V	Vctrl was swept from 0 V to 1 V.	Satisfied
Supply voltage 2-5 V	Could not be verified with supply op-amp because ahdLib opamp failed to compile.	Not fully satisfied in final simulation

15. Conclusion

A voltage-controlled positive phase shifter was successfully designed and simulated using a first-order active all-pass filter structure. The final circuit used a VCVS as the active amplifier and a gpdk90 NMOS as a voltage-controlled resistor. The circuit produced a phase shift from approximately 1.18 degree to 90.66 degree at 10 kHz when Vctrl was swept from 0 V to 1 V.

The gain stayed very close to 0 dB, so the circuit works as an all-pass phase shifter. Transient simulation confirmed that at Vctrl = 0 V, Vin and Vout are almost in phase, and at Vctrl = 1 V, Vout leads Vin by about one quarter cycle, which is close to 90 degree at 10 kHz.

The main limitation is that the 2 V to 5 V supply voltage requirement was not fully verified because the available ahdLib opamp failed to compile in the Cadence environment. Therefore, the final result verifies the phase-shifting operation and control-voltage behavior, while the supply voltage part remains a limitation of this simulation setup.

Appendix A: Final Design Summary

Parameter	Final value
Circuit type	Positive first-order active all-pass phase shifter
Active element	VCVS, gain = 100k
NMOS model	gpdk90 nmos1v
NMOS W	1.2 um
NMOS L	1 um
R1	100 kilo-ohm
R2	100 kilo-ohm
C0	100 pF
Rseries	153 kilo-ohm
Rbias	100e6 ohm
Vctrl range	0 V to 1 V
Final phase range at 10 kHz	about 1.18 degree to 90.66 degree